



GeoBikeLLM: LLM-Powered Bike Route Planning with Air Quality and Vegetation Aware Geospatial Intelligence

A Human-Centric GeoAI System for Healthier, Greener Urban Mobility

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Outline

- Introduction
 - The Problem
 - Why Now?
- Introducing GeoBikeLLM a smarter, healthier way to navigate
 - Case Study
 - How It Works
- Results and Discussion
 - Results & Trade-offs.
 - Why GeoBikeLLM Stands Out.
- Summary & future research
 - Challenges & Future Work.
 - Conclusion: Navigation that cares.

Outline

- Introduction
 - Background & Motivations
 - Spatial data challenges & requirements
 - Spatial join
- A method for simplifying spatial join
 - Overview
 - Approximate spatial join
- Results and Discussion
 - Deployment: baselines & testing setup
 - Approximate spatial join Vs. baseline
- Summary & future research

The Problem – Why Environment-Aware Routing?



You're biking to be healthy... but what if your route is making you **sick**?



- Key Stats:

- 90% of people worldwide breathe polluted air (**WHO**)
- Cyclists are exposed to 2–3× more PM_{2.5} than car drivers **due to proximity to traffic and higher respiration rates**
- Non-exhaust emissions (tire, brake, road dust) now rival tailpipe pollution (**EIT Urban Mobility**)



The Problem – Why Environment-Aware Routing?



Cycling is healthy... but what if your route exposes you to **pollution**?



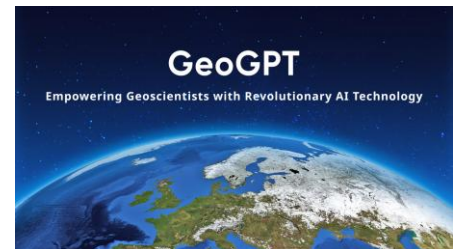
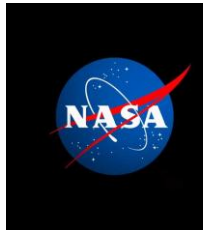
- The Gap
 - Traditional navigation apps (e.g., Google Maps) prioritize **shortest/fastest** routes.
 - They ignore air quality (PM2.5) and green space (NDVI), key factors for cyclist health and experience.
- In cities like **Chicago**, pollution can vary street-by-street (cite Google's Air View project).

Why Now? The Rise of Hyperlocal Environmental Intelligence

-  Invisible Threats Are Now Visible, Thanks to Mobile Sensing & AI
- **Google's Project Air View** has mapped over 500 million air quality measurements across cities like:
 - London, Oakland, Dublin, Houston, Copenhagen, Amsterdam, Hamburg, Bengaluru
 - Equipped Street View cars with sensors to measure PM_{2.5}, NO₂, Black Carbon, O₃ at street level
 - Data used by city planners to:
 - Create low-emission zones
 - Protect schools and playgrounds
 - Inform climate and health policies
- Low-cost sensors (e.g., CitySense, Aclima) now deliver lab-grade accuracy — enabling scalable urban monitoring
- IBM-NASA Prithvi: Open foundation models for satellite-based environmental analysis
- GeoAI explosion: LLMs like GeoGPT, OmniGeo can now reason about space and environment
-  Opportunity: Leverage these advances to build personalized, environment-aware mobility tools — like GeoBikeLLM.



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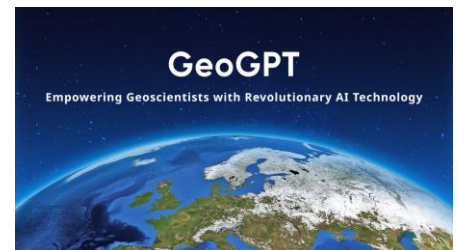
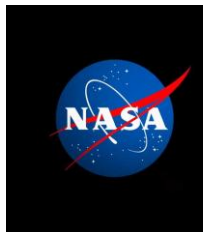


Why Now? The Rise of Hyperlocal Environmental Intelligence

-  Invisible Threats Are Now Visible, Thanks to Mobile Sensing & AI
 - Google's Project Air View: 500M+ street-level measurements in 8+ cities.
 - Low-cost sensors (LCS): Democratizing air monitoring in underserved areas.
 - IBM-NASA Prithvi & FAST-EO TerraMind: Next-gen foundation models for Earth observation.
 - GeoAI explosion: LLMs like GeoGPT, OmniGeo can now reason about space and environment
 - Autonomous GIS: LLMs now power end-to-end spatial reasoning — from natural language to maps — paving the way for systems like GeoBikeLLM that close the loop from intent to action.
-  Opportunity: Leverage these advances to build personalized, environment-aware mobility tools — like GeoBikeLLM.



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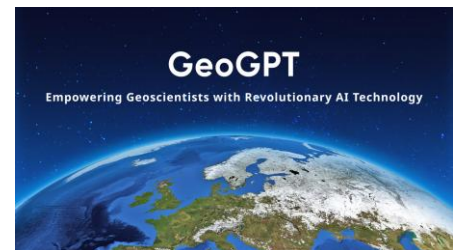
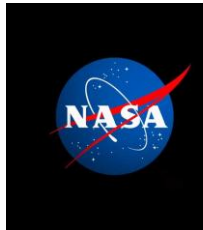
TerraMind

Why Now? The Rise of Hyperlocal Environmental Intelligence

-  Invisible Threats Are Now Visible, Thanks to Mobile Sensing & AI
- IBM-NASA **Prithvi** & FAST-EO **TerraMind**: Next-gen foundation models for **Earth observation (EO)**.



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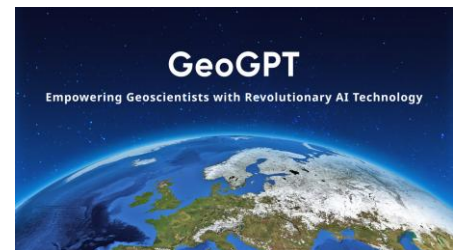
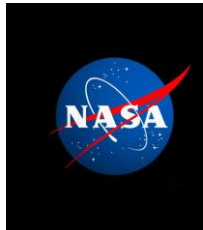


Why Now? The Rise of Hyperlocal Environmental Intelligence

-  Invisible Threats Are Now Visible, Thanks to Mobile Sensing & AI
- Low-cost sensors (LCS): WMO confirms they can democratize air monitoring, especially in underserved areas. Networks in Lagos, London, and Mexico City show their scalability and policy impact.
-  Opportunity: Leverage these advances to build personalized, environment-aware mobility tools — like GeoBikeLLM.



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Why Now? The Rise of Hyperlocal Environmental Intelligence

-  Invisible Threats Are Now Visible, Thanks to Mobile Sensing & AI
- Trends:
 - Mobile sensing networks (e.g., CitySense): Deploy low-cost, vehicle-mounted sensors on taxis and buses to capture hyperlocal, real-time air quality data across urban areas.
- Opportunity:
Leverage these advances to build personalized, **environment-aware mobility tools**.



Google
Geospatial Reasoning?



Our Vision – A Smarter, Healthier Way to Navigate

Healthy & Green Route Finder with Explanation

Travel Prompt

e.g., I want to go from Melrose Park to Hyde Park

Air Quality Penalty (Preference for Healthier Routes – Less Pollution)

0.1

0 1

Greenness Reward (Preference for Greener Routes – Scenic Vegetation)

0.05

0 1

ETA

Explanation for Route Choice

Generate Route

Use via API  • Built with Gradio  • Settings 

- Introducing GeoBikeLLM: Where LLMs Meet Sustainable Mobility
- What It Does:
 - ✓ Understands natural language: *“Take me from Navy Pier to Hyde Park with the cleanest air”*
 - ✓ Balances distance, PM2.5, and NDVI (greenery) in real time
 - ✓ Explains route choices in plain English
 - ✓ Empowers cyclists to prioritize health and well-being

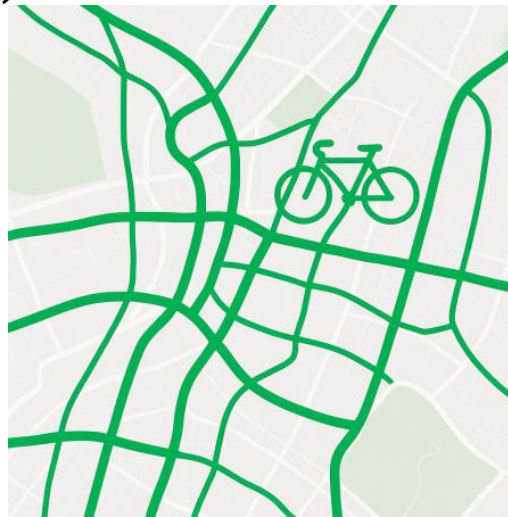
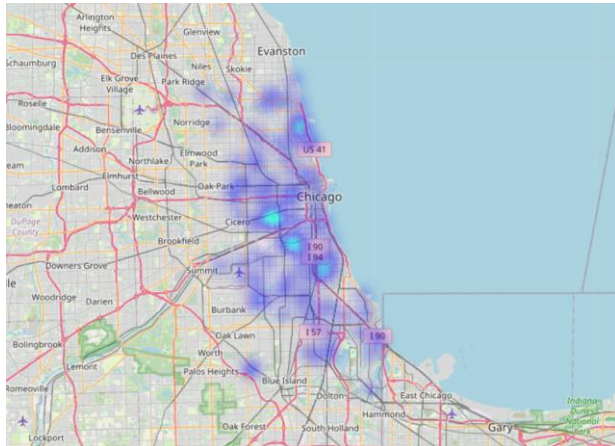
Our Vision – A Smarter, Healthier Way to Navigate

- Why It Matters:
Not just a routing tool — a step toward environmental justice in urban mobility.

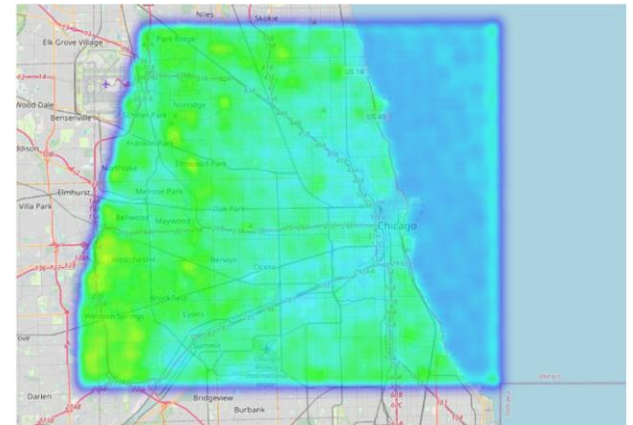
Case Study – Chicago: A City of Contrasts

- Chicago: A Perfect Testbed for Environmental Inequality
- Why Chicago?
 - Diverse urban fabric: dense downtown, industrial zones, green suburbs
 - Rich open data: Eclipse Air Quality Dataset (PM2.5), NDVI, OSM bike network
 - Spatial disparities: PM2.5 varies by 3× across neighborhoods (Google Air View)

PM2.5 —
pollution
hotspots



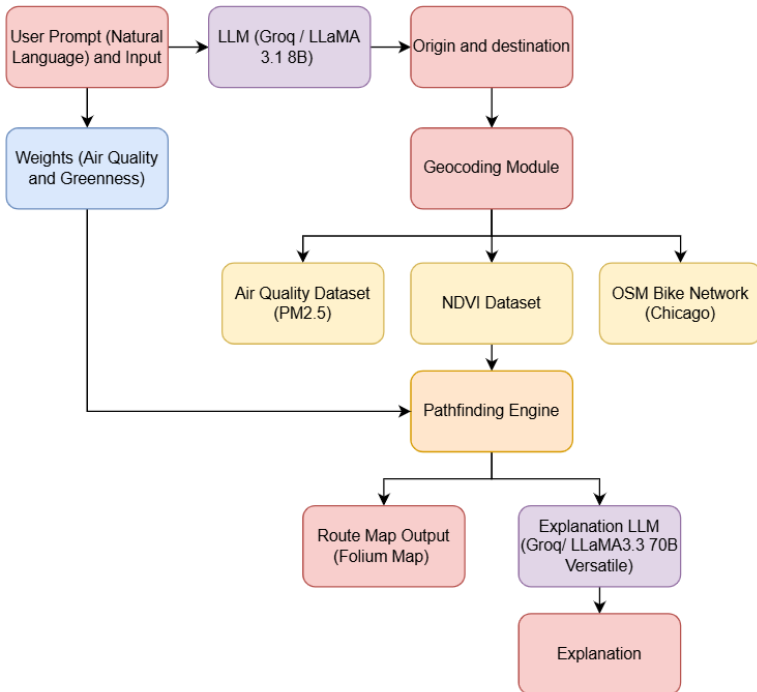
NDVI —
greenery areas



Enabling Data – The Role of Mobile Sensing

- To power environment-aware navigation, we need rich, real-world data.
- Fixed sensors are limited in coverage.
- Mobile platforms like CitySense fill the gap by:
 - Collecting PM2.5, CO2, VOCs from taxis, buses, garbage trucks
 - Providing high-resolution spatial-temporal data
 - Enabling real-time and historical pollution mapping
-  GeoBikeLLM can leverage such systems for accurate, scalable environmental intelligence.

How It Works – From Language to Route



- Two-Stage LLM + Geospatial Optimization
- System Flow (Fig. 1):
 - User says: **“I want to go from Melrose Park to Hyde Park with the greenest route”**
 - LLM 1 (LLaMA 3.1 8B): Extracts origin, destination, preference → JSON
 - Spatial Matching: KD-Tree links route segments to PM2.5 and NDVI
 - Pathfinding: Yen’s K-Shortest Paths with cost:

$$\text{Cost} = \text{length} \times (1 + \alpha \cdot \text{PM2.5} - \beta \cdot \text{NDVI})$$
 - LLM 2 (LLaMA 3.3 70B): Explains why the route was chosen

***Note:** PM2.5 data can be sourced from mobile sensing platforms like CitySense, which provide reliable, high-resolution urban air quality measurements.*

***Note:** Future versions can use IBM-NASA Prithvi-EO for satellite-derived, real-time NDVI.*

Natural Language + Sliders = Robust, Flexible Control

Hybrid Input: Best of Both
Worlds

Two Input Modes:

Natural Language: For
intuitive, conversational
interaction

*“Cleanest air, please”,
“Most trees”, “Fast
but green”*

Sliders: For precise control
over α (air quality) and β
(greenness)

 **Healthy & Green Route Finder with Explanation**

Travel Prompt

e.g., I want to go from Melrose Park to Hyde Park

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Natural Language + Sliders = Robust, Flexible Control

- Why This Matters:

- LLMs can misinterpret ambiguous phrasing (e.g., “green plz” → shortest?)
- Sliders give fine-grained, real-time control
- Users can explore trade-offs interactively

Evaluation – LLM Accuracy & Robustness

Example prompt → parsed JSON

Extract origin, destination, and route preference from the user's travel request.
Valid preferences: "shortest", "healthiest", or "greenest" or combination of multiple of these eg. "shortest and healthiest", "shortest and healthiest and greenest". Handle typos and missing spaces if present, especially in locations names of Chicago City. Please note that this system is designed for Chicago, Illinois, USA only and strictly. Note that healthiest related to air quality and greenest related to Scenic Vegetation.
Return ONLY the result in valid JSON format, with no additional text or explanation.

```
{  
  "origin": "origin name",  
  "destination": "destination name",  
  "preference": "shortest | healthiest | greenest | healthiest and  
shortest | shortest and greenest | healthiest and greenest | shortest  
and healthiest and greenest "  
}
```

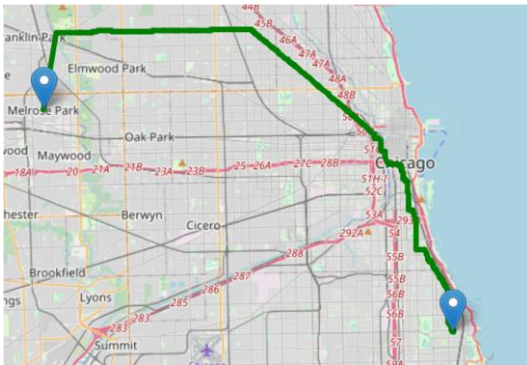
Structured system prompt used to extract origin, destination, and routing preference from natural language input

- 94.3% Accuracy in Understanding Real-World Prompts
- Test: 35 user-generated inputs
 - Typos: “logansqr”, “musuem campus”
 - Informal: “i wanna go frm lakeview to the field musuem green plz”
 - Mixed: “shortest and healthiest”
- Results:
 - 94.3% accuracy in both location and preference extraction.
 - Only 2 failures due to ambiguous phrasing

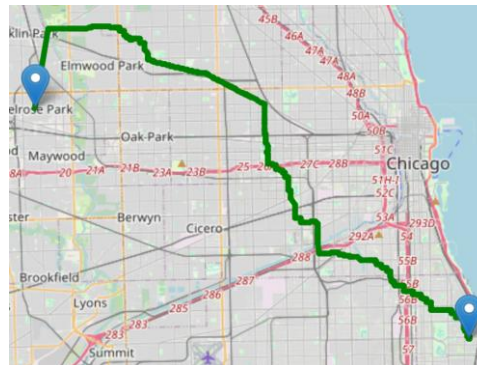
Results – Trade-offs in Action



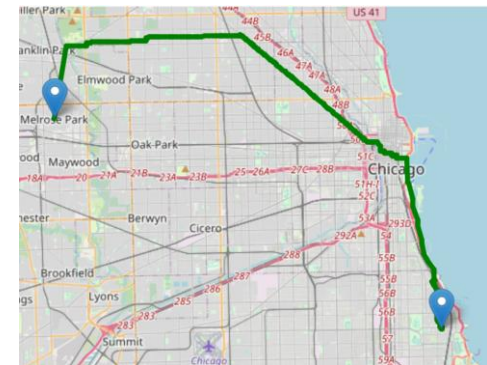
- Three Routes, Three Priorities: Seeing the Trade-offs
- Case: Melrose Park → Hyde Park
 - Route 1 ($\alpha=0, \beta=0$): 32.78 km, 131 min – shortest
 - Route 2 ($\alpha=1, \beta=1$): 34.98 km, 139.9 min – balanced (cleaner + greener)
 - Route 3 ($\alpha=0, \beta=1$): 33.67 km, 134.7 min – greener, scenic



Route 1



Route 2



Route 3

Why GeoBikeLLM Stands Out

First System to Combine LLMs + Environment + Route Planning

A Focused Solution for a Critical Urban Challenge

Contribution:

- ✓ First to close the loop: Natural language → environmental routing → explanation
 - ✓ Inspired by the Autonomous GIS vision: systems that self-generate, self-execute, and self-explain spatial workflows
- GeoBikeLLM automates **intent-to-route** pipeline — a practical step toward human-in-the-loop autonomous GIS

Why GeoBikeLLM Stands Out

- A Step Towards Multimodal GeoAI for Human-Centric Applications

A Focused Solution for a Critical Urban Challenge

System	NLP	Env. Aware	Route Planning	Personalization
ChatGeoAI [7]	✓	X	X	X
GeoQAMap [10]	✓	X	X	X
OmniGeo [15]	✓	✓	X	X
GeoBikeLLM	✓	✓	✓	✓

Contribution:

✓ First to close the loop: Natural language → environmental routing → explanation

We integrate text (LLM), imagery (NDVI), and sensor data (PM2.5) to solve a critical urban health challenge.

A practical, human-centric application of the GeoAI revolution.

Designed for real-world impact, not just analysis.

Why GeoBikeLLM Stands Out

First System to Combine LLMs + Environment + Route Planning

A Focused Solution for a Critical Urban Challenge

SYSTEM	NLP	ENV. AWARE	ROUTE PLANNING	PERSONALIZATION	USER FOCUS
ChatGeoAI[7]	✓	×	×	×	Analysts
GeoQAMap[10]	✓	×	×	×	Researchers
OmniGeo[15]	✓	✓	×	×	Developers
Chat2Geo	✓	✓	×	✓	Experts
GeoBikeLLM	✓	✓	✓	✓	Cyclists

While platforms like Chat2Geo offer powerful, general-purpose geospatial analysis using LLMs and Google Earth Engine, GeoBikeLLM is purpose-built for one critical task: environment-aware bike routing.

A practical, human-centric application of the GeoAI revolution.

Why GeoBikeLLM Stands Out

A Step Toward Autonomous, Human-Centric GeoAI

SYSTEM	NLP	ENV. AWARE	ROUTE PLANNING	PERSONALIZATION	USER FOCUS
ChatGeoAI[7]	✓	×	×	×	Analysts
GeoQAMap[10]	✓	×	×	×	Researchers
OmniGeo[15]	✓	✓	×	×	Developers
Chat2Geo	✓	✓	×	✓	Experts
GeoBikeLLM	✓	✓	✓	✓	Cyclists

Challenges & Future Work

- Scaling to the World: Challenges Ahead
- Challenges:
 - Real-time data: Can we use low-cost sensors or IBM-NASA Prithvi (FM) for live updates?
 - Generalization: Will it work in cities with sparse data?
 - Ethics: Could green routes steer users into unsafe areas?

Challenges & Future Work

- Scaling to the World: Challenges Ahead
- Future Work:
 - Integrate Prithvi-EO (or FAST-EO TerraMind) for better NDVI
 - Add elevation, surface quality, non-exhaust emissions
 - Mobile app + user studies
 - Integrate real-time environmental data streams from mobile sensing platforms (e.g., CitySense) for dynamic, live-updated route recommendations.
 - including LCS networks (e.g., AirQo, Breathe London), which WMO recognizes as key tools for democratizing air quality data.
 - Expand to cities in the Global South using LCS networks (e.g., Lagos, Nairobi, Indore), where WMO-backed initiatives are already building capacity.



TerraMind

Challenges & Future Work

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Conclusion – Navigation That Cares

GeoBikeLLM: Bridging LLMs
and Sustainable Mobility

Takeaways:

- ✓ Natural language + sliders = intuitive, robust interaction
- ✓ Environment-aware routing improves cyclist health and experience
- ✓ LLMs enhance accessibility, transparency, and personalization
- ✓ A scalable model for human-centric GeoAI



GeoBikeLLM is not just a prototype — it's part of an emerging multimodal AI stack, where frameworks like TerraTorch and systems like vLLM are making it possible to fuse LLMs with Earth observation at scale.

Conclusion – Navigation That Cares

- GeoBikeLLM: Bridging LLMs and Sustainable Mobility
- Call to Action:
 - Let's build navigation systems that care not just about *distance*, but about *well-being*.
 - Initiatives like the TGI-AWS (Taylor Geospatial Institute (TGI) and Amazon Web Services (AWS)) **Generative AI Challenge** are opening doors for researchers to turn prototypes like **GeoBikeLLM** into real-world solutions. We're ready to answer that call.
- Vision: A scalable model for human-centric GeoAI.
 - GeoBikeLLM is part of the GeoAI revolution: LLMs + GeoFMs making **Earth intelligence** accessible to everyone.
 - GeoBikeLLM is a step toward Autonomous GIS — where LLMs don't just answer questions, but take action: understanding needs, analyzing Earth data, and delivering personalized, explainable decisions.

Our Solution – GeoBikeLLM

- Introducing GeoBikeLLM: Natural Language Meets Environment-Aware Routing

Key Features:

- ✓ Natural language input (e.g., “**Take me from Navy Pier to Hyde Park with the cleanest air**”)
- ✓ Real-time sliders for air quality and greenness preferences
- ✓ Integrates PM2.5, NDVI, and OpenStreetMap bike network
- ✓ Powered by LLMs for intent extraction and explanation

GeoBikeLLM in the Autonomous GIS Ecosystem

- User says: “Take me somewhere green with clean air”
- ↓
- LLM (LLaMA 3) understands intent → extracts origin, destination, preference
- ↓
- Spatial matching: Links OSM routes to PM2.5 & NDVI via KD-tree
- ↓
- Optimization: Yen’s algorithm with environmental cost function
- ↓
- LLM explains: “This route has 30% more trees and 18% lower PM2.5”
- ↓
- Output: Healthy, explainable route — all with minimal human intervention

From Foundation Models to Human-Centric Apps

- [User Input]
- ↓
- Natural Language → LLM (LLaMA 3)
- ↓
- Intent Extraction → GeoBikeLLM Core
- ↓
- Environmental Scoring → Powered by:
 - PM2.5: Mobile sensing (CitySense, AirQo)
 - NDVI: Satellite models (Prithvi-EO, Clay)
- ↓
- Fine-tuned via: TerraTorch 1.0 (HPO, GeoBench, PEFT)
- ↓
- Deployed via: vLLM (multimodal inference)
- ↓
- [Output: Healthy, explainable route]

Q&A and Contacts

Thanks for your attention!

Question's time...

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